

THE ANALYSIS OF COVARIANCE IN RANDOMIZATION THEORY

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Key words: Analysis of covariance; finite population; randomization theory.

Summary: An analysis of covariance for the finite model is developed on the basis of a linear combination between experimental variate (y) and covariate (x) means in such a fashion that treatment effects are estimated with minimum variance in the situation where the regression coefficient is known. Possible estimators of the regression coefficient are evaluated and the related F-test approximations to the permutation test are examined. The results are compared to the analysis of variance for the finite situation as well as to the usual analysis of covariance for a sample from an infinite population with a normal distribution. The different experimental situations to which each of the analyses of covariance are applicable are contrasted and compared.

1. INTRODUCTION

The importance of the randomization or finite population theory in experimental inference is widely recognized. In contrast to the analysis of variance situation, where knowledge of the finite situation is generally judged to be satisfactory, it seems that in covariance analysis the only work extant is that of Cox (1956) and Robinson (1973). Cox suggested the use of weighted randomization to overcome bias due to the estimation of the regression coefficient. However, in some situations proper weights may be unknown. Additionally this procedure is at a disadvantage should an ordinary analysis of variance be

decided on, since the Yatesian unbiasedness of a design will be destroyed. (A proof will be given in the body of the text.) Indeed it is the close connection in logic, method and aim between analysis of variance and covariance procedures that justify the enquiry of this paper into the analysis of covariance in the situation where the randomization theory is applicable. Robinson's (1973) approach is entirely based on asymptotic arguments, whereas some of the methods of this paper may be of use with small or medium sized samples. However, all comparable results obtained here agree asymptotically with Robinson, as they should.

In the analysis of variance situation the error structure induced by randomization is of such a nature that the simple least squares estimates are also best linear unbiased estimates, as follows from Zyskind (1969) and his references. In the analysis of covariance situation the problem is that the covariate is a random variable from which it follows that a Gauss-Markov approach to estimation is inapplicable. Hence the approach in the first part of the paper is an adaptation of the sampling survey theory approach to linear regression (see Cochran, 1977), where the same problem exists. The inescapable random nature of the covariate enforces a somewhat ad hoc approach to the estimation problem, in which it is necessary to investigate almost all estimators that come to mind.

In the second half of the paper the weights are as suggested by the conventional covariance procedure, so that the natural development is analogous to infinite theory instead of sampling survey theory. As the two approaches appear to offer complementary insights it was decided to retain both, especially as the derivation of some formulas seems more natural by one method rather than the other.

To save space in unnecessary description the general approach and notation will be close to that of Kempthorne (1952, 1955), except that like Cox (1956) experimental unit effects will be

denoted by the variable z , so as to save x for its customary role in the analysis of covariance. Additionally the results of the completely randomized and Latin square designs will be given in appendices, with minimum discussion, and the equations will have numerals attached that correspond to similar equations for randomized blocks.

2. RANDOMIZED BLOCKS

(a) *Mathematical description*

Denote the yield with treatment $k (=1, 2, \dots, t)$ on plot $j (=1, 2, \dots, t)$ of block $i (=1, 2, \dots, r)$ by y_{ijk} . Assume additivity of treatment effects, so that $y_{ijk} = z_{ij} + t_k$, where z_{ij} and t_k are the contributions of plots and treatments respectively, assuming for convenience $\sum_k t_k = 0$. One does not observe the yield of each treatment k on plot j of block i , but merely the yield of treatment k on a randomly chosen plot in block i . Hence the derived model is

$$(1) \quad y_{ik} = \mu + b_i + t_k + \sum_j \delta_{ij}^k (z_{ij} - z_{i.}),$$

where y_{ik} is the yield of treatment k in block i and where δ_{ij}^k is a design random variable equal to one if treatment k falls on plot j in block i , with $\delta_{ij}^k = 0$ otherwise. For a more complete treatment of randomized blocks, Kempthorne (1952) should be consulted. Equation (1) is equivalent to his (4) in Chap. 8, with $e_{ij} = z_{ij} - z_{i.}$

From (1) it is possible to obtain the variance of a treatment mean as

$$\text{var}(y_{.k}) = \sum_{ij} (z_{ij} - z_{i.})^2 / r^2 t = (t-1) \sigma_z^2 / rt,$$

say.

Assume now that there exists an x_{ij} , like the yield of a plot in a uniformity trial during a previous season, by which a plot effect z_{ij} can be estimated. An arbitrary β_0 then gives the usual covariance adjustment

$$\hat{t}_k = y_{.k} - y_{..} - \beta_0(x_{.k} - x_{..}) .$$

From the variance of a treatment mean one obtains

$$\begin{aligned} \text{var}(\hat{t}_k) = & [\Sigma_{ij}(z_{ij} - z_{i.})^2 + \beta_0^2 \Sigma_{ij}(x_{ij} - x_{i.})^2 \\ & - 2\beta_0 \Sigma_{ij}(x_{ij} - x_{i.})(z_{ij} - z_{i.})] / rt , \end{aligned}$$

which is a minimum for

$$(2) \quad \beta_0 = \beta = \Sigma_{ij}(x_{ij} - x_{i.})(z_{ij} - z_{i.}) / \Sigma_{ij}(x_{ij} - x_{i.})^2 .$$

β , corresponds, of course, to the usual least squares estimator for the situation

$$(3) \quad (z_{ij} - z_{i.}) = \beta(x_{ij} - x_{i.}) + g_{ij} ,$$

where g_{ij} represents deviation from regression. Hence for $\beta_0 = \beta$ the minimum variance is given by

$$(4) \quad \text{var}(\hat{t}_k) = \Sigma_{ij} g_{ij}^2 / r^2 t = (t - 1) \sigma_g^2 / rt ,$$

say.

The important thing to note is that, by the derivation of (2), a reasonable definition of the regression coefficient for the finite model was achieved. It is, furthermore, important that no properties, such as it being a random variable, are assumed for the g_{ij} ; e.g. the derivation of (4) follows from the design random variables defined in (1).

In the usual experimental situation β is unknown, since z_{ij} is unobservable. Hence the question of the estimation of β needs to be dealt with.

(b) *Estimators of β*

As the plot effects z_{ij} are not observable, it follows that functions of the experimental responses y_{ik} need to be used for estimation purposes. Denote the sums of squares and products by B and R for between treatments and residuals, respectively, and define $T = B + R$. The variables involved are dealt with by subscripts, e.g. R_{xy} is equal to the residual sum of products between x and y .

(i) Eliminating treatments effects

Denote

$$(5) \quad \hat{\beta} = \frac{R_{xy}}{R_{xx}}$$

$$= \frac{T_{xz} - B_{xz}}{T_{xx} - B_{xx}},$$

where the z - s can be substituted for the y - s, since the treatment effects cancel.

Since $B_{xz} = \beta B_{xx} + B_{gx}$ it follows from (2) that

$$(6) \quad \hat{\beta} = \beta - B_{gx}/R_{xx}$$

$$= \beta - \varepsilon_1, \text{ say.}$$

From (3) and its associated least squares equations $T_{gx} = 0$, so that $E(\hat{\beta}) = \beta$, approximately, since the expected values

of sums of products are analogous to those of sums of squares. Analogous to the variance of the treatment sum of squares given by Kempthorne (1952), for homogeneous error variances between blocks,

$$\text{var}(B_{gx}) = 2(r-1)T_{gg}T_{xx}/(t-1)r^3.$$

Hence it follows from (6) and the variance of a ratio that

$$\begin{aligned}\text{var}(\hat{\beta}) &= \{2/(t-1)r(r-1)\}T_{gg}/T_{xx} \\ (7) \quad &= 2\sigma_g^2/(r-1)T_{xx}.\end{aligned}$$

(ii) Ignoring treatments effects

Denote

$$(8) \quad \tilde{\beta} = T_{xy}/T_{xx}$$

from which it follows by (1) that

$$\begin{aligned}(9) \quad \tilde{\beta} &= \beta' + r\sum_k t_k(x_{.k} - x_{..})/T_{xx} \\ &= \beta + \varepsilon_2, \quad \text{say.}\end{aligned}$$

Since x_{ik} does not contain treatment effects, the expectation of a treatment mean is equal to the grand mean.

Hence $E(\tilde{\beta}) = \beta$.

From the variances and covariances between treatment means (from Kempthorne, 1952) one obtains

$$\begin{aligned}(10) \quad \text{var}(\tilde{\beta}) &= \sum_k t_k^2/(t-1)\sum_{ij}(x_{ij} - x_{i.})^2 \\ &= \sigma_t^2/T_{xx}, \quad \text{say.}\end{aligned}$$

- (iii) Constructing an unbiased estimator eliminating treatment effects

In a comparison between (5) and (8) it may be of value to distinguish between the possible effects of bias and variability. Hence define

$$(11) \quad \bar{\beta} = \{r/(r-1)\} R_{xy}/T_{xx},$$

so that, by analogy to the variance of the treatment sum of squares given by Kempthorne (1952), it follows that

$$(12) \quad \begin{aligned} \text{var}(\bar{\beta}) &= \{2/(t-1)r(r-1)\} \{\beta^2 + T_{gg}/T_{xx}\} \\ &= 2\sigma_z^2/(r-1)T_{xx}. \end{aligned}$$

(c) *First moments of sums of squares*

- (i) The conventional analysis of covariance

From (5), (6), (9) and (10) it follows that

$$E(T_{yy} - \tilde{\beta}^2 T_{xx}) = T_{gg} + r \sum_k t_k^2 - E(\epsilon_2^2) T_{xx},$$

and that

$$E(R_{yy} - \hat{\beta}^2 R_{xx}) = E(R_{gg} - \epsilon_1^2 R_{xx}).$$

Hence, the conventional formulae (e.g. Kempthorne, 1952) and (7) and (10) give approximately

$$(13) \quad \left\{ \begin{aligned} &E\{\text{sum of squares for treatments adjusted}/(t-1)\} \\ &= \{1 + 2/r(t-1)\} \sigma_g^2 - \{1/(t-1) - r\} \sigma_t^2 \\ &\text{and that} \\ &E\{\text{sum of squares for error}/(r-1)(t-1)\} \\ &= \{1 - 2/r(r-1)(t-1)\} \sigma_g^2. \end{aligned} \right.$$

It is, therefore, clear that the conventional analysis of covariance is only asymptotically unbiased in the Yatesian sense, as both equations must have the same right-hand sides, under the hypothesis of no treatment effects, to be unbiased. This biased nature of the infinite procedure in the finite situation suggests that alternatives be looked for. From Section a it is clear that, if β is known, the corrected treatment means have variances of precisely the same form as in the ordinary analysis of variance situation. Furthermore, $\tilde{\beta} = \beta$, under the hypothesis of no treatment effects. These facts suggest that there may be sense in analyzing $w_{ik} = y_{ik} - \beta x_{ik}$, and to follow the ordinary analysis of variance procedures on

$$(14) \left\{ \begin{array}{l} \hat{w}_{ik} = y_{ik} - \hat{\beta} x_{ik} \\ \text{or} \\ \tilde{w}_{ik} = y_{ik} - \tilde{\beta} x_{ik} \\ \text{or} \\ \bar{w}_{ik} = y_{ik} - \bar{\beta} x_{ik} . \end{array} \right.$$

(ii) Analysis of variance, correcting with $\hat{\beta}$

Using the same relationships as in (i) it follows, on analyzing $\hat{w}_{ik} = y_{ik} - \hat{\beta} x_{ik}$ in an ordinary analysis of variance, that

$$(15) \left\{ \begin{array}{l} E(\text{treatments ms}) = \{1 + 2/(t-1)(r-1) + 2/(t-1)r\}\sigma_g^2 + r\sigma_t^2 \\ \text{and that} \\ E(\text{error ms}) = \{1 - 2/r(r-1)(t-1)\}\sigma_g^2 . \end{array} \right.$$

(iii) Analysis of variance, correcting with $\tilde{\beta}$

Use of $\tilde{w}_{ik} = y_{ik} - \tilde{\beta}x_{ik}$, (9) and (10) in an analysis of variance results, exactly, in

$$(16) \quad \left\{ \begin{array}{l} E(\text{treatments ms}) = \sigma_g^2 + \{1/r(t-1) + 2(r-1)/r^2(t-1)^2 - 2/(t-1) + r\} \sigma_t^2 \\ \text{and} \\ E(\text{error ms}) = \sigma_g^2 + \{1/r(t-1) - 2/r^2(t-1)^2\} \sigma_t^2, \end{array} \right.$$

under the assumption of homogeneous variances in the variable x for the different blocks. In the situation of heterogeneous variances, a correction to the terms with $r^2(t-1)^2$ as denominator is applicable. This correction is the same as the one applied by Kempthorne (1952, p. 142 equation (15)) to the factor $2(r-1)/r^3(t-1)$, since both have a common origin.

Note that this procedure provides an exact unbiased test in the Yatesian sense.

(iv) Analysis of variance, correcting with $\bar{\beta}$

Assume homogeneous variances in the variables x and z for the different blocks. Then use of (11), (12) and the technique of (25) gives exactly

$$(17) \quad \left\{ \begin{array}{l} E(\text{treatments ms}) = \{(1+4/(t-1)r)\} \sigma_g^2 + \{2/r(r-1)(t-1) \\ \quad - 8(r-2)/r^2(r-1)(t-1)^2\} \sigma_z^2 + r \sigma_t^2 \\ \text{and} \\ E(\text{error ms}) = \{1-4/(t-1)r(r-1)\} \sigma_g^2 + \{2/r(r-1)^2(t-1) \\ \quad + 8(r-2)/r^2(r-1)^2(t-1)^2\} \sigma_z^2. \end{array} \right.$$

Note the appearance of both σ_g^2 and σ_z^2 .

(d) *Second moments of sums of squares*

Under the null hypotheses of no treatment effects it follows from (6) and (7) that $\text{var}(\text{treatments adjusted}) = \text{var}\{B_{gg} + \epsilon_1^2 R_{xx}\}$ and $\text{var}(\text{treatments} + \text{error}) = 0$, for the analysis of covariance. Approximate the fourth moment of the treatment sum of squares, given by Pitman (1937), for both r and t large, by $12\{(r-1)(t-1)/r\}^2 \sigma_g^8$. Hence the variance of the beta ratio is approximately equal to

$$(18) \quad \{2(r-1)(t-1)/r + 16(r-2)/r^2 + 8/r^2\} \sigma_g^4 / T_{gg}^2,$$

for both r and t large, as follows by using the same sort of argument on the covariance term as in the derivation of (25). In ordinary analysis of variance the term $16(r-2)/r^2 + 8/r^2$ is excluded from the numerator of (18).

For the analysis of variance on $\hat{w}_{ik} = y_{ik} - \hat{\beta}x_{ik}$ the beta distribution analog of the F-test gives

$$\frac{B_{ww}^{\wedge}}{T_{ww}^{\wedge}} = (B_{gg} + \epsilon_1^2 T_{xx} + \epsilon_1^2 R_{xx}) / (T_{gg} + \epsilon_1^2 T_{xx}).$$

The variance of $B_{ww}^{\wedge}/T_{ww}^{\wedge}$ is difficult to get in any sort of reasonable form, but the binomial expansion of the denominator suggests (18) as an approximate value also in this situation. Unfortunately it appears that the variance of $B_{ww}^{\wedge}/T_{ww}^{\wedge}$ has a poorer approximation to the analysis of variance situation than (18). Under the null hypothesis, the procedure involving $\tilde{\beta}$ gives, of course, the same result as in ordinary analysis of variance.

The reasonably close approximation of the analysis of variance procedures involving $y_{ik} - \hat{\beta}x_{ik}$ to the first moments of the finite model suggests the approximate variance of $y_{.k} - \hat{\beta}(x_{.k} - \bar{x}_{.})$ for large sample sizes to be equal to $\{(t-1)/t\} \sigma_g^2 / r$, from (4), where the term $(t-1)/t$ can be ignored for the purposes

of comparison between treatments. An estimate of σ_g^2 can be obtained from (13), or alternatively (15) or (16). For smallish sample sizes the variance obtained by dividing $E(\text{treatments})$ in (15) by r , neglecting the term $r\sigma_t^2$ is recommended. The variance for comparisons between treatment effects is approximately equal to

$$(19) \quad \{1/r + 2/(t-1)r(r-1) + 2/(t-1)r^2\}\sigma_g^2.$$

When $\tilde{\beta}$ is used to obtain corrected means, the mean squared error of a treatment effect is exactly equal to

$$\sigma_g^2/r + \{1/r^2(t-1) + 2(r-1)/r^3(t-1)^2\}\sigma_t^2,$$

with the same possible correction to the last term, as in the situation of (16).

The bias in corrected treatment means when $\hat{\beta}$ or $\bar{\beta}$ is used is $O\{r^3(t-1)\}^{-\frac{1}{2}}$, and when $\tilde{\beta}$ is used is equal to $-t_k/r(t-1)$, for the k -th corrected treatment mean. Hence for $\tilde{\beta}$ the variance of a treatment effect is exactly equal to

$$(20) \quad \sigma_g^2/r + \{(rt-2)/r^3(t-1)^2\}\sigma_t^2$$

with a term, $2\{\Sigma(T_i - \bar{T})^2 / \bar{T}^2 r^4(t-1)^2\}\sigma_t^2$,

that needs to be subtracted with heterogeneous within block variances in x , where $T_i = \Sigma_j (x_{ij} - x_{i.})^2$ and where $\bar{T} = \Sigma_i T_i / r$. When $\bar{\beta}$ is used, the mean squared error of a corrected treatment mean is exactly equal to

$$(21) \quad \{1/r + 4/r^2(t-1)\}\sigma_g^2 + [\{2r(t-1) - 8(r-2)\}/r^3(r-1)(t-1)^2]\sigma_z^2.$$

As in (17), note again the appearance of both σ_g^2 and σ_z^2 . Note furthermore the similarity of the σ_g^2 -term with (19), suggesting that (19) may be a reasonably close approximation to its true value.

For the purpose of comparison with (19), the mean variance of an estimate of a treatment effect in infinite analysis of covariance can be shown to be equal to

$$\{1/r + 1/(t-1)r(r-1)\}\sigma^2.$$

The errors involved in the approximations associated with the treatment mean squares in (13) and (15) are $O(1/r^2)$ and in the residual mean squares of (13) and (15) together with (19) the errors are $O(1/r^3)$, as follows from the variances given by Pitman (1937).

3. WEIGHTED AND UNWEIGHTED RANDOMIZATION

(a) *Weighted randomization and the analysis of variance*

The process of weighted randomization proposed by Cox (1956) is to select each experimental arrangement with probability proportional to R_{xx} . Then, if f is any function of the observations x and y , its expectation under weighted randomization is

$$(22) \quad E_W(f) = E_P(wf)/E_P(w),$$

where E_W and E_P represent expectations over weighted and unweighted randomizations respectively.

Let $w = R_{xx}$ and $f = R_{yy}$ for randomized blocks.

Then, from Kempthorne (1952) $E_P(f) = (t-1)(r-1)\sigma_Z^2$, where $\sigma_Z^2 = \sum_{ij} (z_{ij} - z_{i.})^2 / r(t-1)$, according to (1).

By symmetry it follows from Kempthorne (1952, p. 141) that, under the null hypothesis,

$$E_P(R_{xx}R_{yy}) = \{2(t-1)(r-1)/r\}\sigma_x^2\sigma_Z^2 + \{(t-1)(r-1)\}^2\sigma_x^2\sigma_Z^2,$$

so that

$$(23) \quad E_W(R_{yy}) = \{(2/r + (t-1)(r-1))\sigma_z^2\},$$

from (22).

In the same way it follows that

$$E_P(R_{xx}B_{yy}) = \{-2(r-1)(t-1)/r + (r-1)(t-1)^2\sigma_x^2\sigma_z^2\},$$

so that

$$(24) \quad E_W(B_{yy}) = \{-2/r + (t-1)\}\sigma_z^2.$$

It follows clearly from (23) and (24) that weighted randomization causes the ordinary analysis of variance procedure to be biased in the Yatesian sense. This may be a disadvantage since the analysis of covariance is often regarded as an optional procedure, to be adopted only if it gives a great enough improvement on the analysis of variance.

The effect of weighted randomization on the variance of B_{yy} follows from the identity

$$\begin{aligned} E\{R_{xx}\{B_{yy} - E(B_{yy})\}^2\} &= E\{R_{xx} - E(R_{xx})\}\{B_{yy} - E(B_{yy})\}^2 \\ &\quad + E(R_{xx})E\{B_{yy} - E(B_{yy})\}^2, \end{aligned}$$

and from the third moment of B_{yy} which follows from Pitman (1937) as equal to $\{8(r-1)(r-2)(t-1)/r^2\}\sigma_z^6$.

Hence

$$(25) \quad E\{R_{xx}\{B_{yy} - E(B_{yy})\}^2\}/E(R_{xx}) = \{-8(r-2)/r^2 + 2(r-1)(t-1)/r\}\sigma_z^4,$$

from which it seems fair to conclude that small values of r , $r \neq 2$, may give rise to serious disturbances from the usual

variance of the treatment sum of squares.

(b) *Weighted and unweighted randomization in the analysis of covariance*

Since the answers come out in a somewhat neater form than in other design situations, consider again randomized blocks. The error sum of squares is equal to $R_{yy} - R_{xy}^2/R_{xx}$. Hence from (22) one needs $E_p(R_{xx}R_{yy} - R_{xy}^2)$. In the same way as in the derivation of (23) it follows that

$$(26) \quad E_p(R_{xy}^2) = \{2(t-1)(r-1)/r\}\sigma_x^2\sigma_z^2 + \{(t-1)(r-1)\}^2\sigma_{xz}^2,$$

where $\sigma_{xz} = \Sigma_{ij}(x_{ij} - x_{i.})(z_{ij} - z_{i.})/r(t-1)$, from (3).

Hence, from (23) and (26), it follows immediately that

$$(27) \quad E_w(R_{yy} - R_{xy}^2/R_{xx}) = (t-1)(r-1)(\sigma_z^2 - \sigma_{xz}^2/\sigma_x^2) \\ = (t-1)(r-1)\sigma_g^2,$$

where $\sigma_g^2 = \Sigma_{ij}g_{ij}^2/r(t-1)$, from (3).

Thus a comparison of (27) and (13) shows that weighted randomization, as is also proved by Cox (1956), neatly compensates for the disturbing factor in (13). Hence under weighted randomization it follows that

$$(28) \quad E(\text{treatments adjusted}) = (t-1)\sigma_g^2 - \{1/(t-1) - r\}\Sigma_k t_k^2,$$

as follows from the use of $\tilde{\beta}$ in the sum of squares for treatment plus error.

It follows from (27) and (28) that the same number of degrees of freedom is applicable to both the analyses of variance and

of covariance in the finite situation. As in the derivation of (18) it follows for r and t both large that $\text{var } B_{gx}^2 = 8\{[(r-1)(t-1)/r]\sigma_g^2\sigma_x^2\}^2$, and hence from (18), (22) and (25) that, under weighted randomization, approximately,

$$(29) \quad \text{var}(\text{treatments/adjusted}) = \{2(r-1)(t-1)/r + 8(r-1)/r^2\}\sigma_g^4.$$

Hence weighted randomization also moves the second moment nearer to the analysis of variance situation, but not as much as the first.

The variance of a treatment effect under weighted randomization follows from noting that

$$r\Sigma_k(\hat{w}_{.k} - \hat{w}_{..} - t_k)^2 = B_{gg} + \epsilon_1^2(R_{xx} + B_{xx}) + \epsilon_1^2R_{xx} \quad \text{where} \\ \hat{w}_{ik} = y_{ik} - \hat{\beta}x_{ik}. \quad \text{Hence, from (24), (6) and (7) it follows that, approximately,}$$

$$(30) \quad \text{var}(\hat{t}_k) = \{1/r + 2/(t-1)r(r-1)\}\sigma_g^2,$$

for the comparison of treatment means. The error in (30) is $O(1/r^3)$, as follows from Pitman (1937). The factor 2 on the right-hand side of (30) differs from Cox (1956), who obtains the conventional formula for average treatment effects. The factor 2 arises from permutation sampling variances, and is avoided in infinite theory by conditioning on the observed values of x , a process which is impossible in the finite theory of covariance analysis. This impossibility follows as the stochastic properties of all other variables can be derived from the properties of the design random variables defined in (1).

4. A PRACTICAL EXAMPLE

In view of the individuality of members of cross-fertilizing species it seems that the finite model analysis might be

more appropriate than conventional covariance for a set of data used by Steel and Torrie (1960, p. 326) as an example for multiple covariance. The factorial nature of the treatments was ignored for present purposes. The experiment was concerned with growth (weight gain) of Guinea pigs on Ladino clover on different soil types and fertilization levels. The Guinea pigs were blocked in three groups on initial weight, and analysis of variance shows no significant treatment effects on intake.

Hence the data are analysed with y = weight gain, x = amount of forage consumed, according to a randomized blocks design, with three blocks and eight treatments.

Some quantities of interest are:

$$\hat{\beta} = 0,1398; \quad \hat{s} = 0,026$$

$$\tilde{\beta} = 0,1255; \quad \tilde{s} = 0,064$$

$$\bar{\beta} = 0,1370; \quad \bar{s} = 0,040$$

where s = standard errors of the β .

The standard errors of treatment effects, according to (19, 20, 21 and 30) respectively, are

$$\text{using } \hat{\beta}, \quad \text{s.e.} = 14,16$$

$$\text{using } \tilde{\beta}, \quad \text{s.e.} = 14,53$$

$$\text{using } \bar{\beta}, \quad \text{s.e.} = 14,39$$

$$\text{using } \hat{\beta}, \quad (\text{weighted}) \text{ s.e.} = 13,60$$

$$\text{using conventional s.e.} = 13,34$$

covariance.

These values are calculated from equating (13) to the observed values in the conventional covariance procedure, to obtain $\sigma_g^2 = 485,78$ and $\sigma_t^2 = 2989,18$; $\sigma_z^2 = 1123,50$. The analyses of variance correcting with $\hat{\beta}$, $\tilde{\beta}$ and $\bar{\beta}$ (14) give approximately the same values as the analysis of covariance for the F-test,

an observation which can be explained by the similar values of $\hat{\beta}$, $\tilde{\beta}$ and $\bar{\beta}$.

5. DISCUSSION

1. Often weighted randomization will give the smallest variance for estimates of the treatment effects, but the degrees of freedom for the F-test may have to be adjusted to be a better approximation to the permutation test, according to procedures given by Kendall and Stuart (1976). The approximations necessary for the derivation of the variance of the treatment sum of squares (29) involve, however, an unknown measure of risk, especially with small samples. The additional disadvantages are that the residual sums of squares in the supplementary variable must be known before the experiment, that the randomization procedure is more difficult, and that the associated analysis of variance on the primary variable has a worse approximation of the F-test to the permutation test than with ordinary randomization. For additional remarks Cox (1956) should be consulted.

2. The choice between $\hat{\beta}$ and $\bar{\beta}$ obviously depends on the relative sizes of σ_g^2 and σ_z^2 in (21). That (21) is non-asymptotic should count in favour of $\bar{\beta}$.

3. The analysis of variance on (14), adjusting with a regression coefficient ignoring treatments ($\tilde{\beta}$), may do remarkably well, even with large differences between treatment effects, as is illustrated by the practical example. Its small sample type I properties are identical to those of the ordinary finite analysis of variance. On the other hand it follows from (18) that analysis of variance adjusting with $\hat{\beta}$ or $\bar{\beta}$ and ordinary infinite analysis of covariance have second moments that may not approximate those of the F-test very well. Judging from first moments under H_1 , it seems that there may be problems with lack of power on small samples (r and t both small) connected with

the use of $\tilde{\beta}$. However, analyses of variance on (14) adjusting with a regression coefficient excluding treatments ($\hat{\beta}$ or $\bar{\beta}$) are free from this defect, again judging from first moments. Furthermore, adjustment of the degrees of freedom may be an uncertain remedy, as follows from the approximations involved in the derivation of (18).

The conclusion follows that, with small samples, both the analysis of variance procedures (on 14) together with the ordinary analysis of covariance procedure should be followed, neglecting the adjustment in degrees of freedom for analysis of covariance. The complementary nature of the properties of $\tilde{\beta}$ and $\hat{\beta}$ or $\bar{\beta}$ under both H_0 and H_1 , assures a small possibility of erroneous conclusions with small discrepancies between the F-tests.

In the situation of sizable discrepancies between the F-tests, it may, nevertheless, be advisable to adjust the degrees of freedom of the F-test by the use of (13) and (18), according to the method of Kendall and Stuart (1976). The last resort may be to do an actual permutation test on, say, some function of the adjusted treatment means.

4. In view of the importance of randomization theory in the situation of heterogeneous experimental units, it is clear that a finite model analysis of covariance would be applicable in the case where, in infinite theory, the response on different experimental units would be described by a so-called model of the second kind (vide Fisk, 1967; Nelder, 1968; Sprent, 1969), where a unique relationship is postulated for each experimental unit. It is, however, unnecessary to make any assumptions of linearity or homogeneity in finite analysis of covariance, since the distributional properties of the residuals follow from the randomization procedure.

ACKNOWLEDGEMENT

This paper is dedicated to professor Oscar Kempthorne in appreciation of his leading role in the explication of randomization theory.

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APPENDIX I: THE COMPLETELY RANDOMIZED DESIGN

The equations are numbered according to corresponding equations for randomized blocks. Equations identical to those in randomized blocks are not given. The pertinent reference for the analysis of variance is Kempthorne (1955).

$$(7) \quad \text{var}(\hat{\beta}) = \{2(t-1)(rt-1)/t^2(r-1)^2\}\sigma_g^2/T_{xx}$$

$$(10) \quad \text{var}(\tilde{\beta}) = \{r(t-1)/(rt-1)\}\sigma_t^2/T_{xx}$$

$$(13) \quad \left\{ \begin{array}{l} E\{\text{sum of squares for treatments (adjusted)/(t-1)}\} \\ = \{1 + 2/t(r-1)\}\sigma_g^2 - \{r/(rt-1) - r\}\sigma_t^2 \\ \text{and} \\ E\{\text{sum of squares for error}/t(r-1)\} \\ = \{1 - 2(t-1)/t^2(r-1)^2\}\sigma_g^2. \end{array} \right.$$

ing to corresponding equations for randomized blocks. Let the yield of treatment k on the i -th row and j -th column be represented by $y_{ijk} = z_{ij} + t_k$, so that the error term added to treatment effects (after correcting for rows and columns) is equal to $z_{ij} - z_{i.} - z_{.j} + z_{..} = f_{ij}(z)$, say. Let x_{ij} again be the observable covariate, and also define

$$x_{ij} - x_{i.} - x_{.j} + x_{..} = f_{ij}(x) .$$

Analogous to the previous designs, define

$$(8) \quad \tilde{\beta} = \Sigma_{ij} f_{ij}(y) f_{ij}(x) / \Sigma_{ij} f_{ij}^2(x) ,$$

so that

$$(9) \quad \tilde{\beta} = \beta + t \Sigma_k t_k (x_k - x_{..}) / \Sigma_{ij} f_{ij}^2(x) .$$

From the variance and covariance of treatment means derived by Kempthorne (1952) it follows that

$$(10) \quad \text{var}(\tilde{\beta}) = \{t/(t-1)^2\} \Sigma_k t_k^2 / \Sigma_{ij} f_{ij}^2(x) .$$

Let furthermore

$$f_{ij}(z) = \beta f_{ij}(x) + g_{ij} ,$$

so that $\Sigma_{ij} g_{ij} f_{ij}(x) = 0$. From (9) and (10) it is then possible to derive expectations of the sums of squares in an analysis of variance on $\tilde{w}_{ijk} = y_{ijk} - \tilde{\beta} x_{ijk}$.

It follows, approximately, that

$$(16) \quad \left\{ \begin{array}{l} E(\text{treatments ms}) = \sigma_g^2 - t\{2t/(t-1)^3 - 1\}\sigma_t^2 + 0\{t/(t-1)^3\}\sigma_t^2 \\ \text{and that} \\ E(\text{error ms}) = \sigma_g^2 + 0\{t/(t-1)^3\}\sigma_t^2 . \end{array} \right.$$

From (16) it is now possible to solve for σ_g^2 and to use

$$\{(\tau-1)/t\}\sigma_g^2/t$$

as the estimated large sample variance of an adjusted treatment mean. If a better approximation is desired, the variance for comparisons between treatment means adjusted by $\tilde{\beta}$ is equal to

$$(20) \quad \sigma_g^2/t + \sigma_t^2/(\tau-1)^3 + O\{\sigma_t^2/(\tau-1)^4\},$$

where $\sigma_g^2 = \Sigma_{ij} g_{ij}^2/(\tau-1)^2$ and $\sigma_t^2 = \Sigma_k t_k^2/(\tau-1)$. The bias in an adjusted treatment mean is equal to $-t_k/(\tau-1)^2$. Hence, although a full treatment of Latin squares is impossible, enough results are available to do a finite model analysis of covariance on squares with a reasonably large number of sides.

On the basis of the investigation by Welch (1937) it appears that the normal theory approximation to the randomization theory may be satisfactory for Latin squares of sides of 5 or more. Thus one may expect behaviour analogous to randomized blocks from squares of these sizes. Hence it may be conjectured that (19) and (30) may become useful approximations for $t \geq 5$.

$$(19) \quad \text{var}(\hat{t}_k) = \{1/t + 2/t(\tau-1)(\tau-2) + 2/t(\tau-1)^2\}\sigma_g^2.$$

$$(30) \quad \text{var}(\hat{t}_k) = \{1/t + 2/t(\tau-1)(\tau-2)\}\sigma_g^2.$$